EE 319
Electronics Lab I

EE 319
Lab 01
Presentation

By Steven Placzek and Jeremy Burke

EE 319
Electrical
Engineering
Lab I

Lab 01
Band Pass Filter
Circuit Analysis

By Steven Placzek and Jeremy Burke

Lab 01
Passive Band-Pass Filter
Introduction

The objective of this lab is to analyze the frequency response of a circuit that acts as a passive band-pass filter. The circuit's response will be determined using frequency domain methods, simulated with LTspice (SPICE), and measured using the Analog Discovery Studio.

Lab 01

Analog Discovery Studio Protoboard

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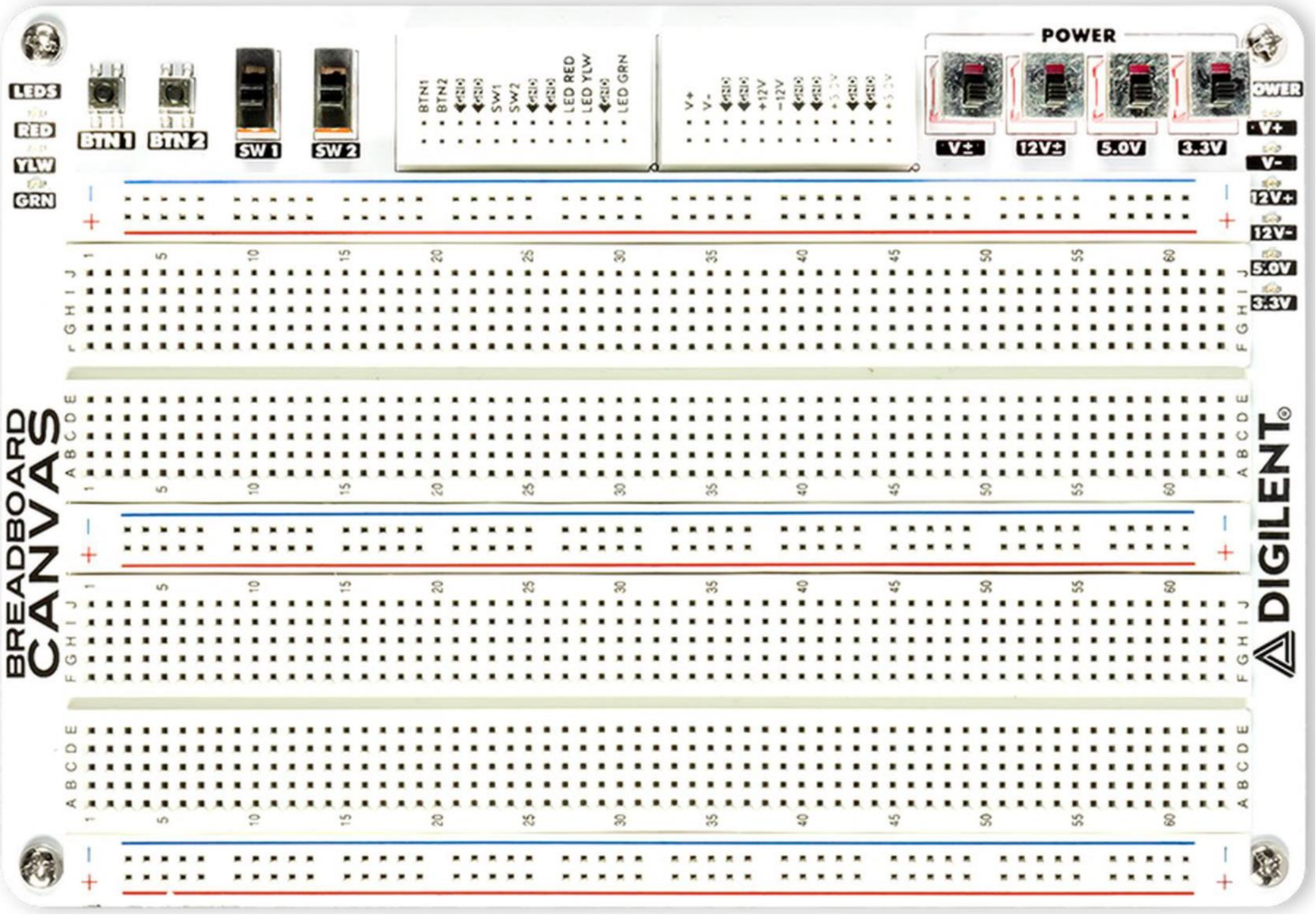
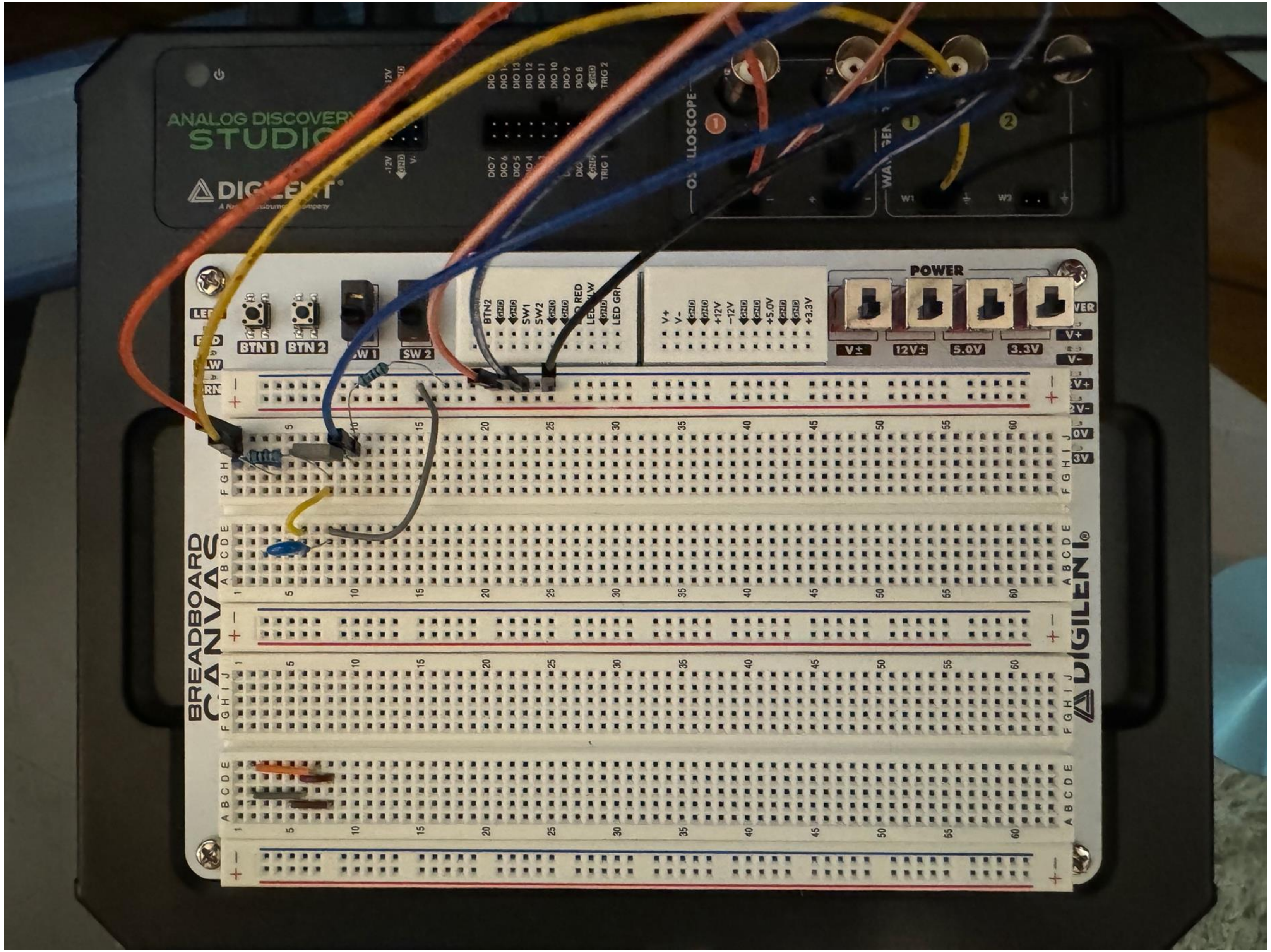


Analog Discovery Studio Protoboard Layout



Lab 01

Protoboard Circuit Layout

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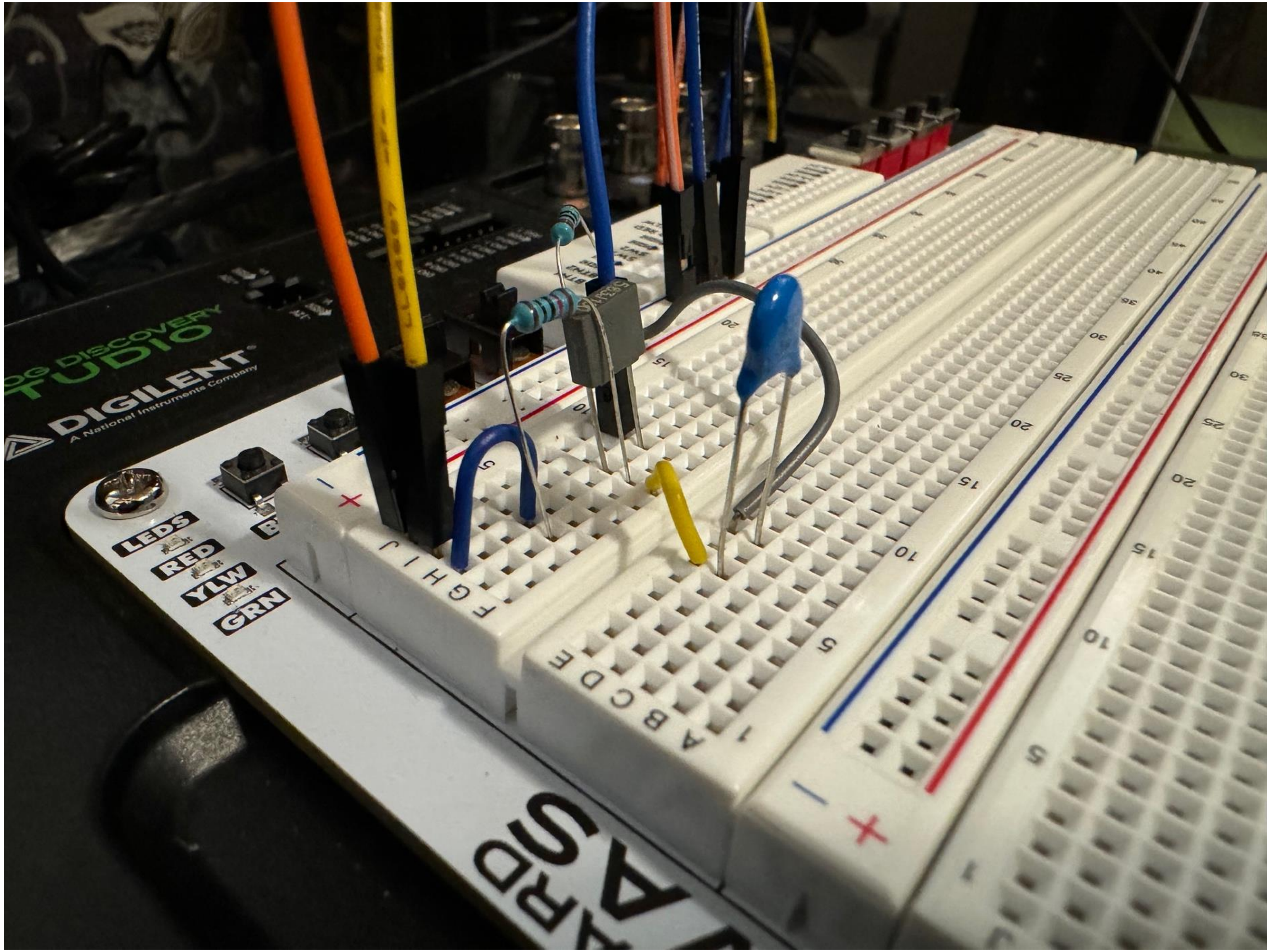
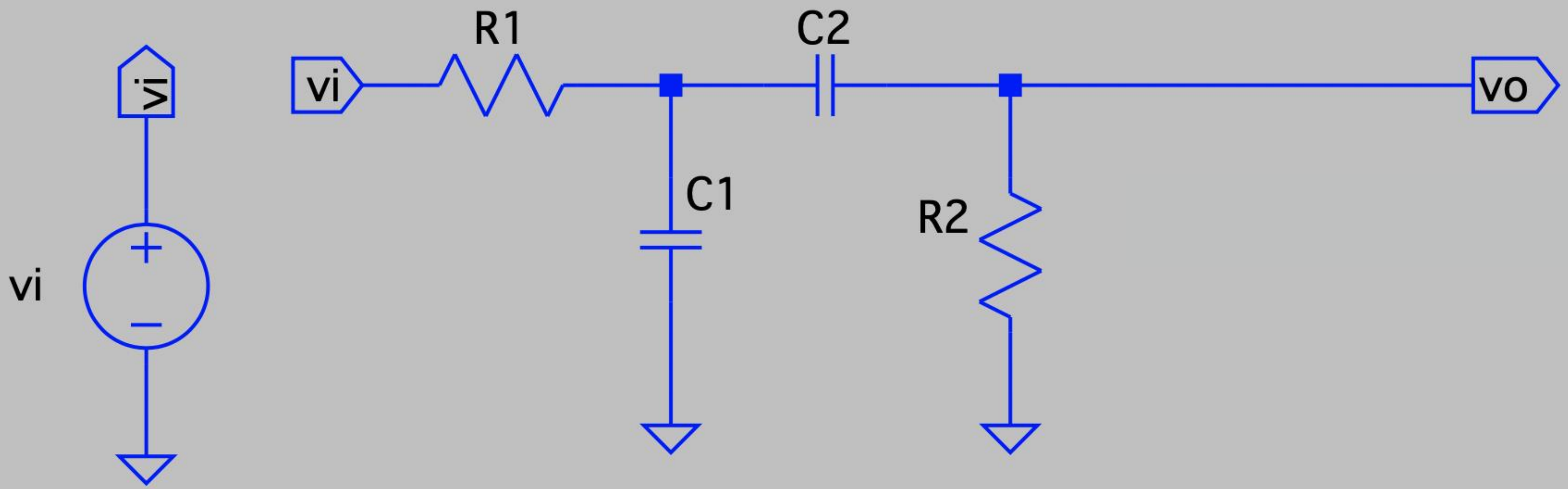


Lab 01 Results

Circuit Diagram with Component Values



$$R_1 = 10\ \mathrm{k\Omega}$$
$$R_2 = 10\ \mathrm{k\Omega}$$
$$C_1 = 56\ \mathrm{nF}$$
$$C_2 = 3.3\ \mathrm{nF}$$

Lab 01 Results

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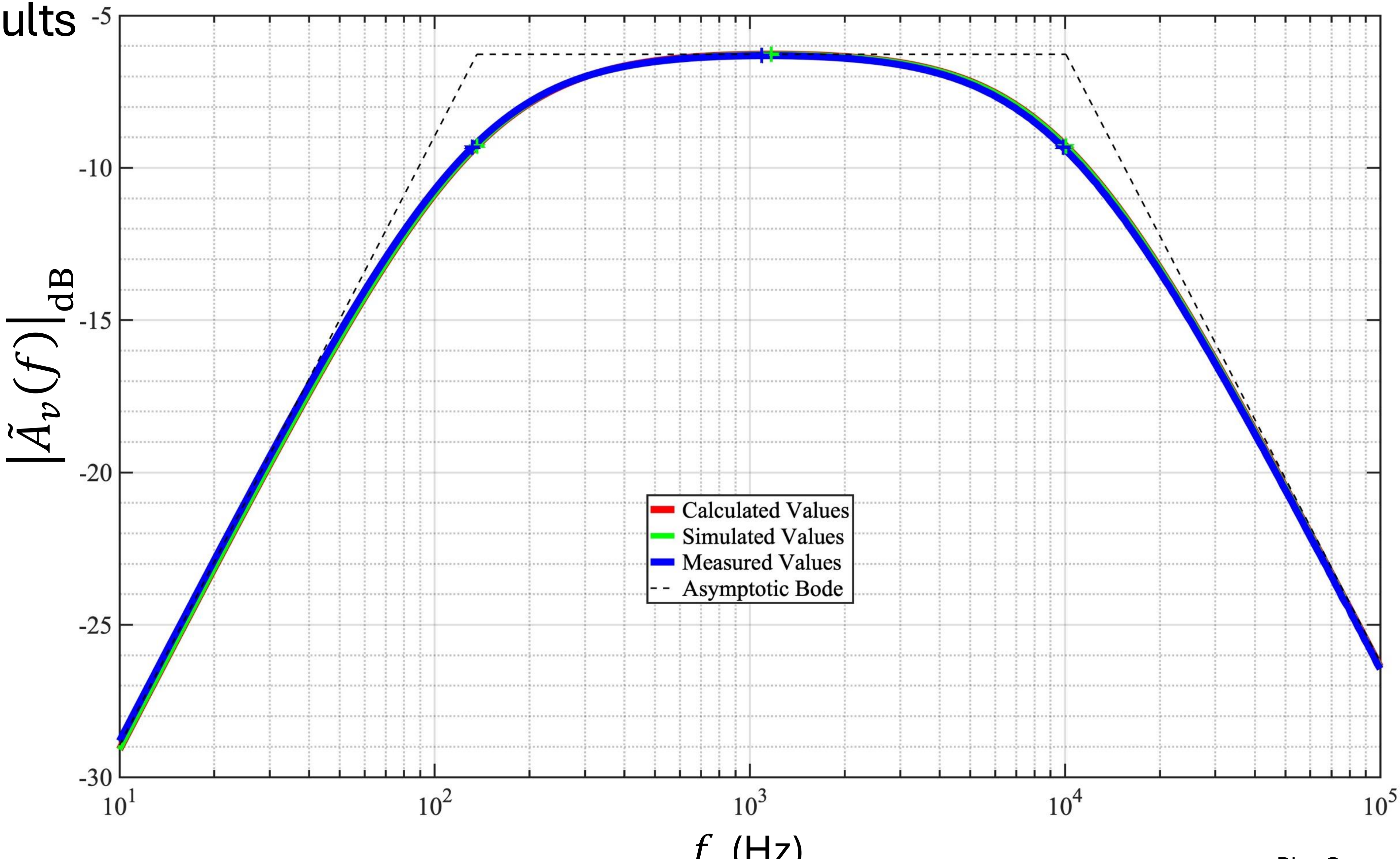
Two Poles: $\omega_L(f) = 140.0096\ \mathrm{Hz}$ & $\omega_H(f) = 9.7899\ \mathrm{kHz}$



Plot Generated using MATLAB

Lab 01 Results

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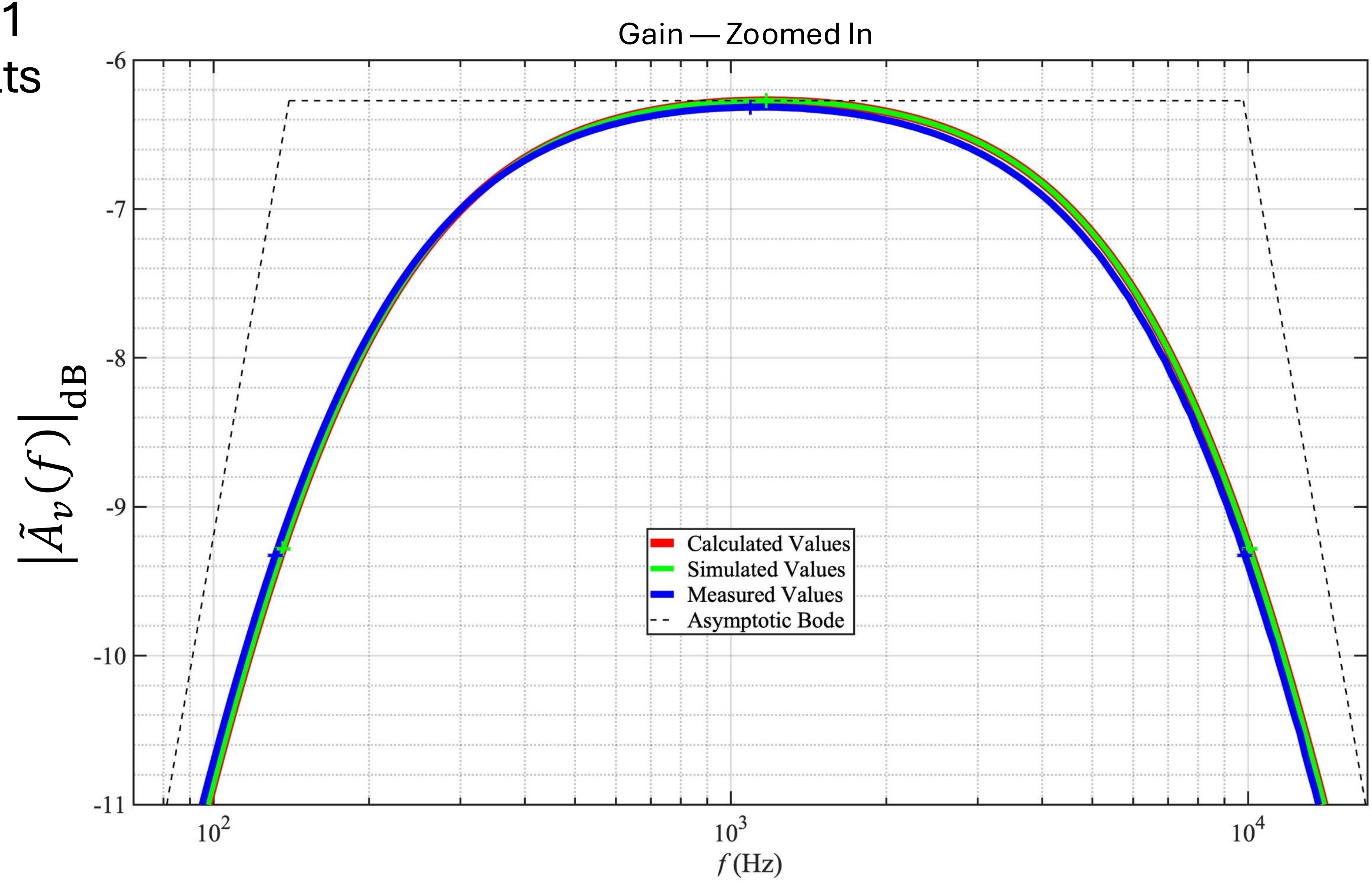

Two Poles: $\omega_L(f) = 140.0096\ \mathrm{Hz}$ & $\omega_H(f) = 9.7899\ \mathrm{kHz}$

Plot Generated using MATLAB

Lab 01
Results

Phase

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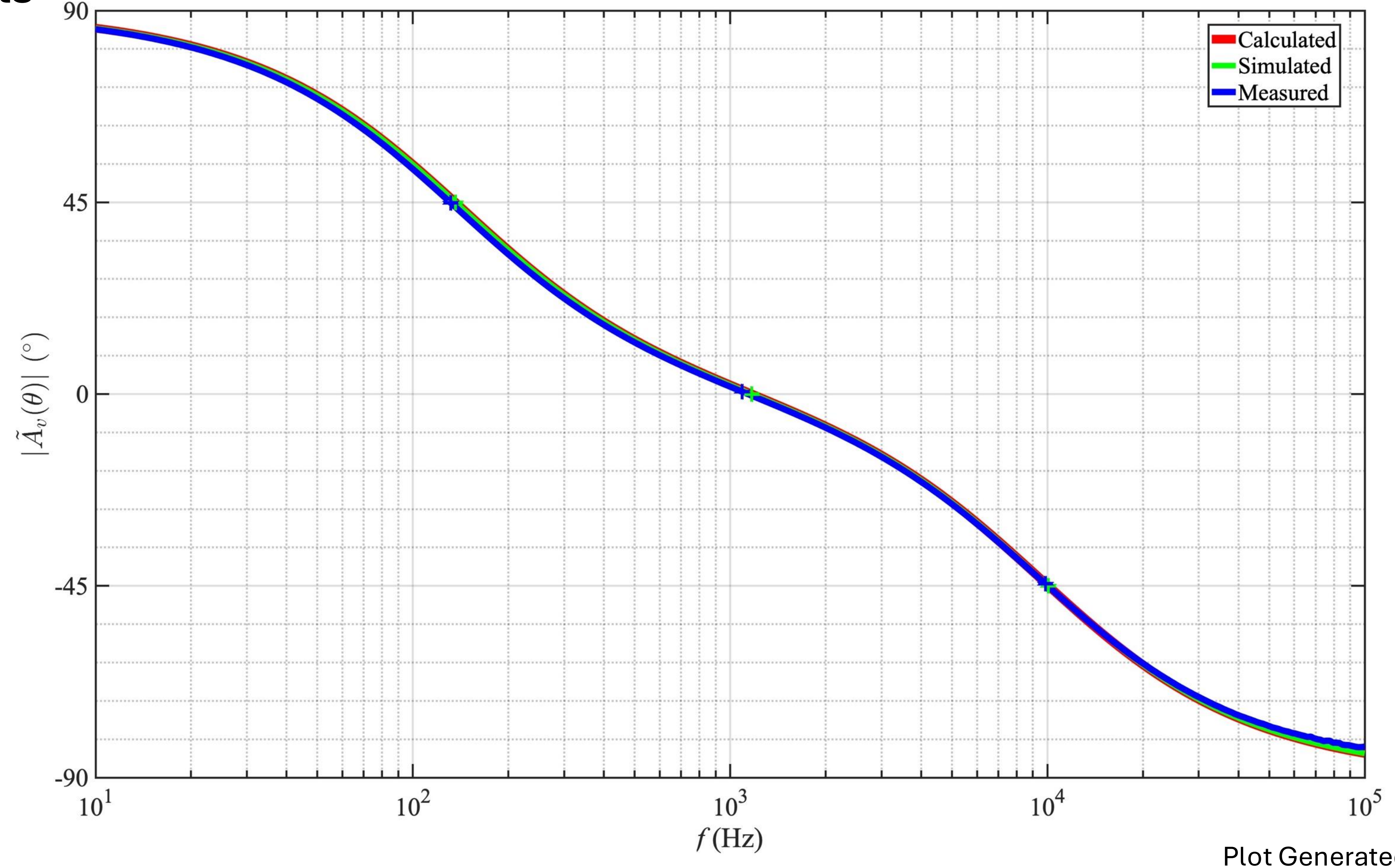


Plot Generated using MATLAB

Calculated, Simulated, & Measured Results

	Calculated	Simulated	Measured	Units
$A_{v(mid)}$	485.6895	485.6895	483.2372	$\mathrm{m}V/V$
$\|A_{v(mid)}\|_{dB}$	-6.2728	-6.2728	-6.3168	dB
f_L	136.1683	136.1683	131.6514	Hz
f_0	1.1708	1.1708	1.0906	kHz
f_H	10.0661	10.0661	9.8537	kHz
BW_f	9.921	9.921	9.7221	kHz
Q	0.1179	0.1179	0.1122	Hz/Hz

Comparison of Calculated & Simulated Values

	Calculated	Simulated	Units	Diff Calc-Sim	% Diff Calc-Sim
$A_{v(mid)}$	485.6895	485.6895	$^{V}/_{V}$	0.0000	0.0000%
$\|A_{v(mid)}\|_{dB}$	-6.2728	-6.2728	dB	0.0000	0.0000%
f_L	136.1683	136.1683	Hz	0.0000	0.0000%
f_0	1.1708	1.1708	kHz	0.0000	0.0000%
f_H	10.0661	10.0661	kHz	0.0000	0.0000%
BW_f	9.921	9.921	kHz	0.0000	0.0000%
Q	0.1179	0.1179	Hz/Hz	0.0000	0.0000%

Comparison of Calculated & Simulated Values

The match between the calculated and simulated responses was nearly identical, with a maximum percentage difference of 0.0000%.

Comparison of Calculated and Measured Results

	Calculated	Measured	Units	Diff Calc-Meas	% Diff Calc-Meas
$A_{v(mid)}$	485.6895	483.2372	V/V	0.0025	0.5062
$\|A_{v(mid)}\|_{dB}$	-6.2728	-6.3168	dB	0.0440	0.6985
f_L	136.1683	131.6514	Hz	4.5168	3.3730
f_0	1.1708	1.0906	kHz	80.1402	**7.0877**
f_H	10.0661	9.8537	kHz	212.4220	2.1328
BW_f	9.921	9.7221	kHz	207.9052	2.1159
Q	0.1179	0.1122	Hz/Hz	0.0057	4.9737

Lab 01 Results

Comparison of Calculated & Measured Values

The alignment between the calculated and measured responses was satisfactory:

- The maximum percent difference was approximately 7.0877% for f_0.
  - The discrepancies between calculated and measured values for f_L, f_0, f_H, and BW_f and measured values are within the components' tolerance limits.

EE 319 Lab 01 – Class Statistics

	Mean	Median	Std Dev	Max	Min	
$A_{v(mid)}$	484.5524	483.8314	3.2280	493.4706	479.5742	mV/V
$\lvert A_{v(mid)} \rvert_{dB}$	-6.2934	-6.3061	0.0577	-6.1348	-6.3829	dB
f_L	135.7873	132.1976	7.2274	152.0204	128.3698	Hz
f_0	1.1933	1.1767	0.0566	1.3160	1.0906	kHz
f_H	10.3822	10.3418	0.4813	11.1313	9.6985	kHz

EE 319 Lab 01 – $A_{v(mid)}$ — Statistical Comparison

Statistic	Class Value (mV/V)	Comparison to Measured Value (483.2362 mV/V)
Mean	484.5524	Lower than mean by 1.3152
Median	483.8314	Lower than median by 0.5942
Std Dev	3.2280	Within 1 standard deviation
Max	493.4706	Lower than max by 10.2334
Min	479.5742	Higher than min by 3.6630

EE 319 Lab 01 – $|A_{v(mid)}|_{dB}$ — Statistical Comparison

Statistic	Class Value	Comparison to Measured Value (-6.3168 dB)
Mean	-6.2934 dB	Measured is lower by 0.0234 dB
Median	-6.3061 dB	Measured is lower by 0.0234 dB
Std Dev	0.0577	Within 1 standard deviation
Max	-6.1348 dB	Measured is lower by 0.182 dB
Min	-6.3829 dB	Measured is higher by 0.0661 dB

EE 319 Lab 01 – f_L — Statistical Comparison

Statistic	Class Value	Comparison to Measured Value (131.6514 Hz)
Mean	135.7873 Hz	Lower than mean by 4.1359 Hz
Median	132.1976 Hz	Lower than median by 0.5462 Hz
Std Dev	7.2274	Within 1 standard deviation
Max	152.0204 Hz	Lower than max by 20.369 Hz
Min	128.3698 Hz	Higher than min by 3.2816 Hz

EE 319 Lab 01 – f_0 — Statistical Comparison

Statistic	Class Value	Comparison to Measured Value (1.0906 kHz)
Mean	1.1933 kHz	Lower than mean by 0.1027 kHz
Median	1.1767 kHz	Lower than median by 0.0861 kHz
Std Dev	7.2274	More than 1 standard deviation lower
Max	1.3160 kHz	Lower than max by 0.2254 kHz
Min	1.0906 kHz	Equal to class minimum

EE 319 Lab 01 – f_H — Statistical Comparison

Statistic	Class Value	Comparison to Measured Value (9.8537 kHz)
Mean	10.3822 kHz	Lower than mean by 0.5285 kHz
Median	10.3418 kHz	Lower than median by 0.4881 kHz
Std Dev	0.4813	Within 1 standard deviation
Max	11.1313 kHz	Lower than max by 1.2776 kHz
Min	9.6985 kHz	Higher than min by 0.1552 kHz

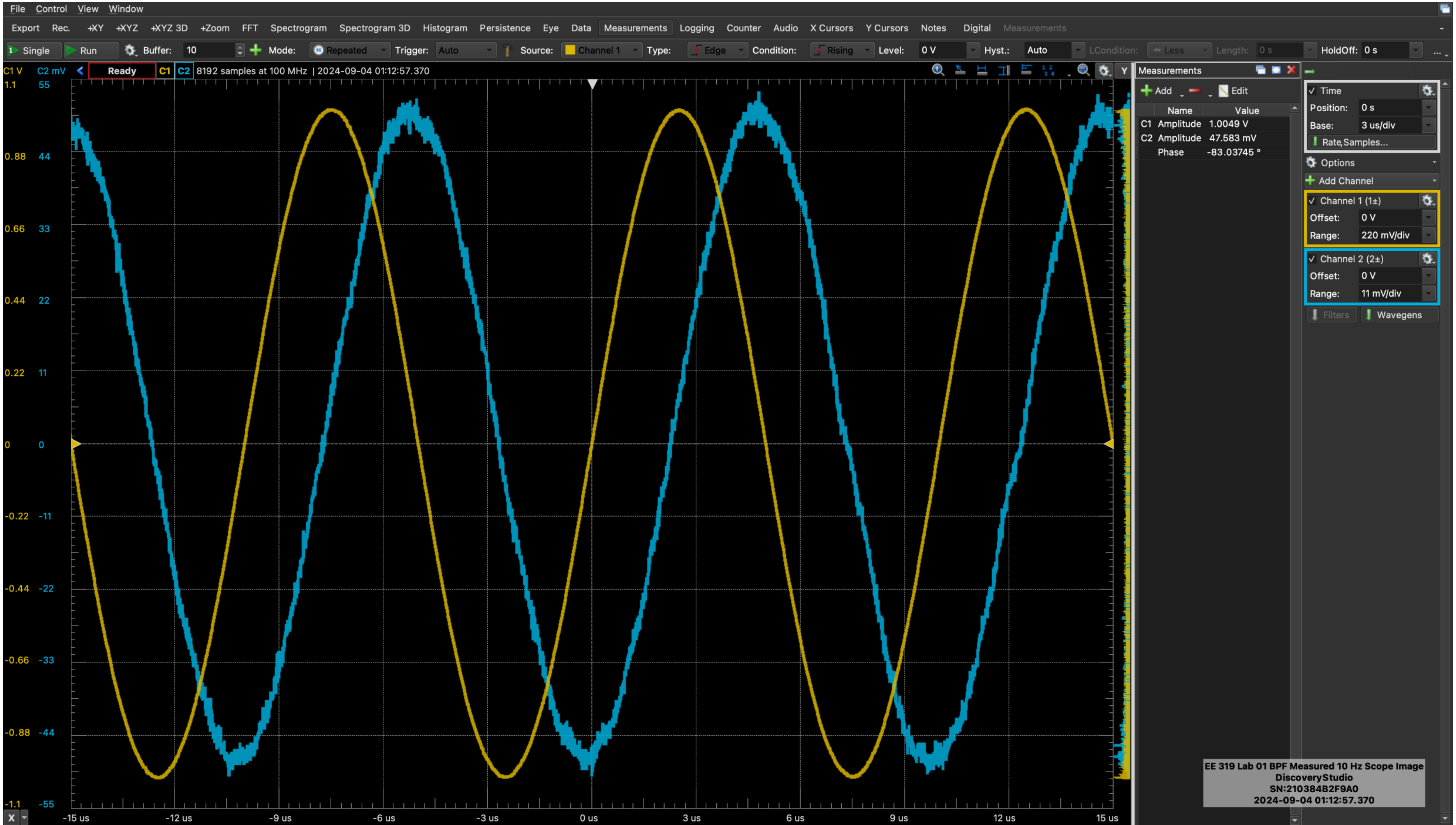
Statistical Analysis

- $A_{v(mid)}$ — The measured value is very close to both the mean and median and falls within 1 standard deviation.
- $|A_{v(mid)}|_{dB}$ — The measured value is also within 1 standard deviation of the class data, which indicates good agreement.
- f_L — The measured value is slightly lower than both the mean and median, but still within 1 standard deviation.
- f_0 — The measured value is lower than the class mean and median and falls outside 1 standard deviation, which suggests it is an outlier in the dataset.
- f_H — This measured value is lower than the class average and median, but still within 1 standard deviation.

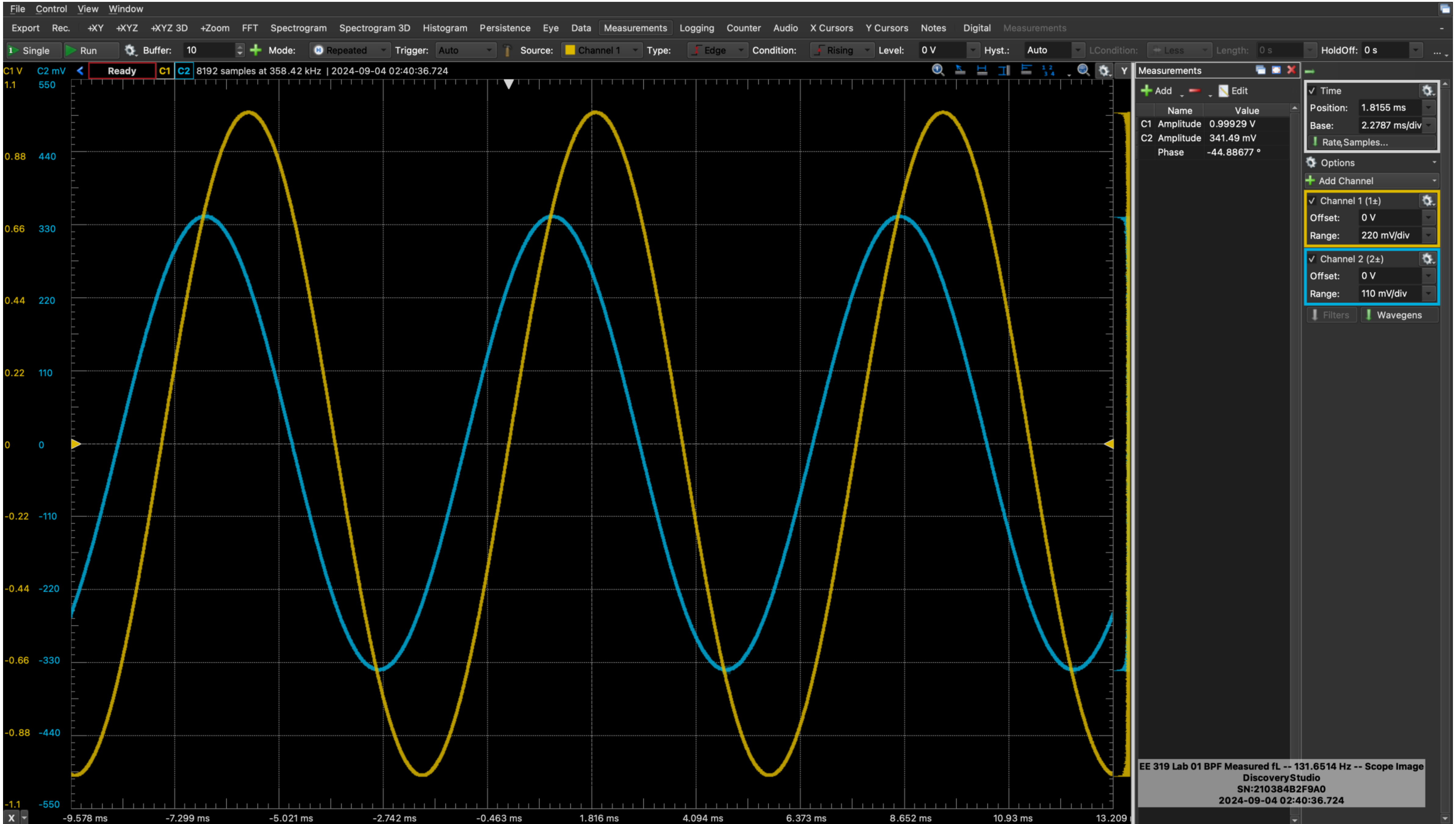
Table of Appendices

Appendix I
Measurements

Oscilloscope at $f = 10$ Hz: $C1 = v_i(t)$ & $C2 = v_o(t)$

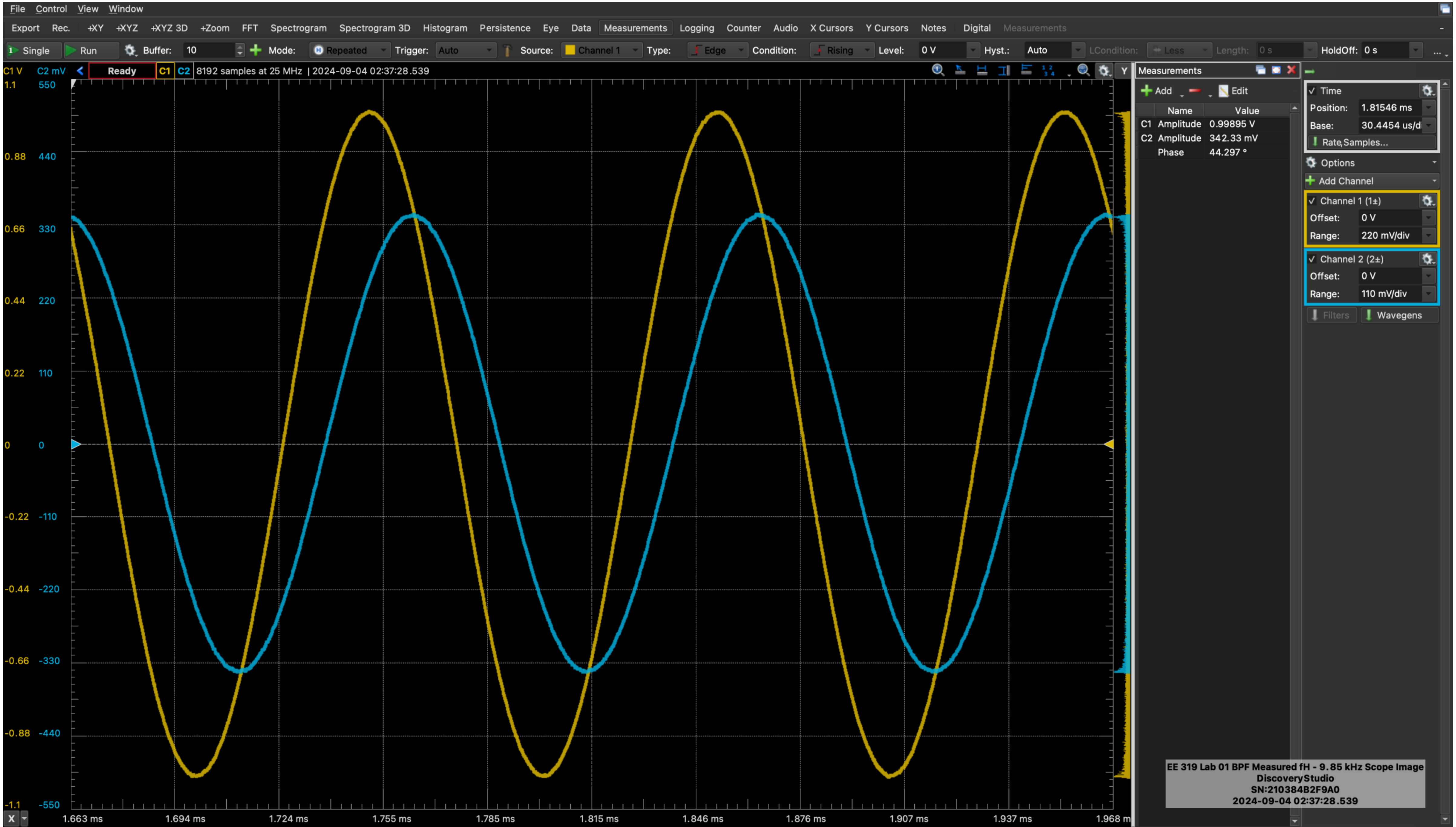


Oscilloscope at $f_{\mathrm{L}} = 131.6514$ Hz: C1 $= v_i(t)$ & C2 $= v_o(t)$

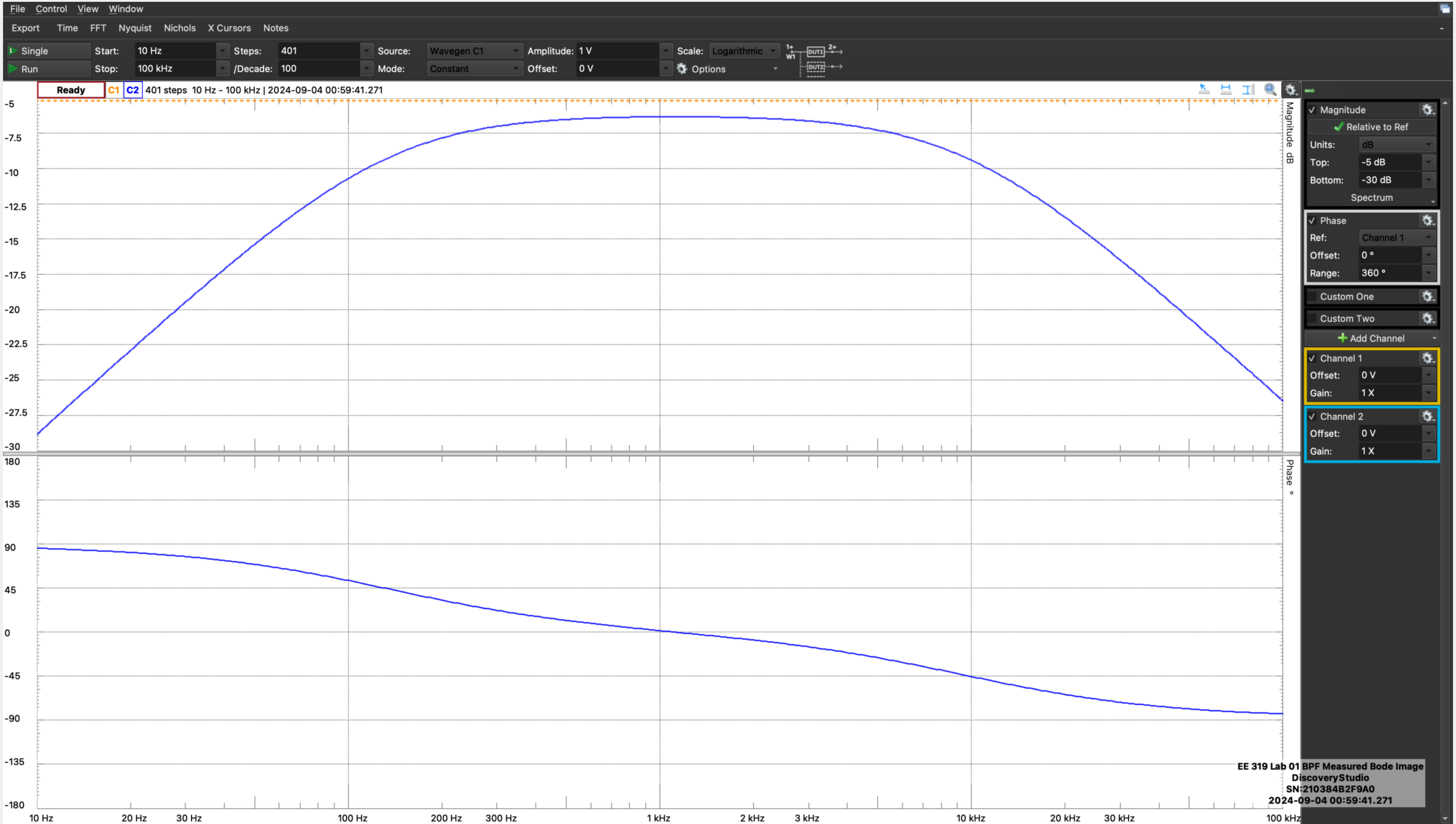


Oscilloscope at $f_0 = 1.0906\ \text{kHz}$: $\text{C1} = v_i(t)$ & $\text{C2} = v_o(t)$

Oscilloscope at $f_{\mathrm{H}} = 9.8537\ \mathrm{kHz}$: $\mathrm{C1} = v_i(t)$ & $\mathrm{C2} = v_o(t)$



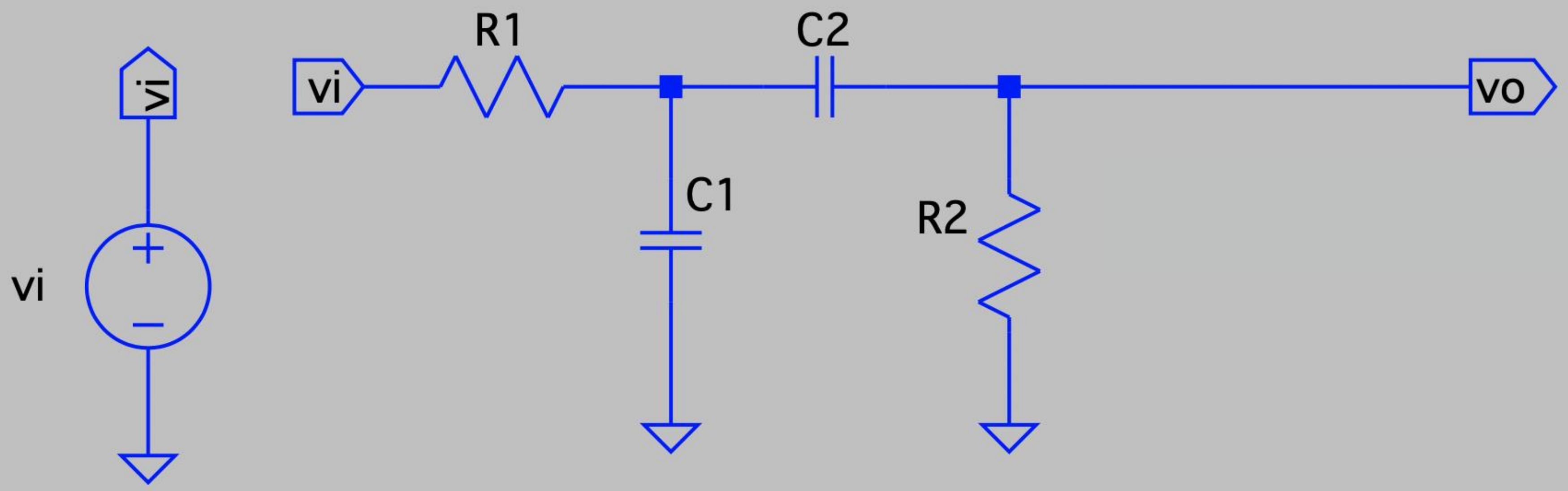
Magnitude & Phase Bode Plots from 10 Hz to 100 kHz



Appendix II
Calculations

Appendix II —Calculations

Circuit Diagram with Component Values



$$R_1 = 2.7\ \mathrm{k\Omega}$$
$$R_2 = 18\ \mathrm{k\Omega}$$
$$C_1 = 100\ \mathrm{nF}$$
$$C_2 = 4.7\ \mathrm{nF}$$

Lab 01

Appendix II — Calculations

$$G_1 = \left(\frac{1}{R_1}\right) \quad G_1 = (100.000 + j0.0000)\ \mu\text{S}$$

$$G_1 = (100.0000 + j0.0000)\ \mu\text{S}$$

$$G_2 = \left(\frac{1}{R_2}\right)$$

$$\omega = 2\pi f$$

$$s = j\omega$$

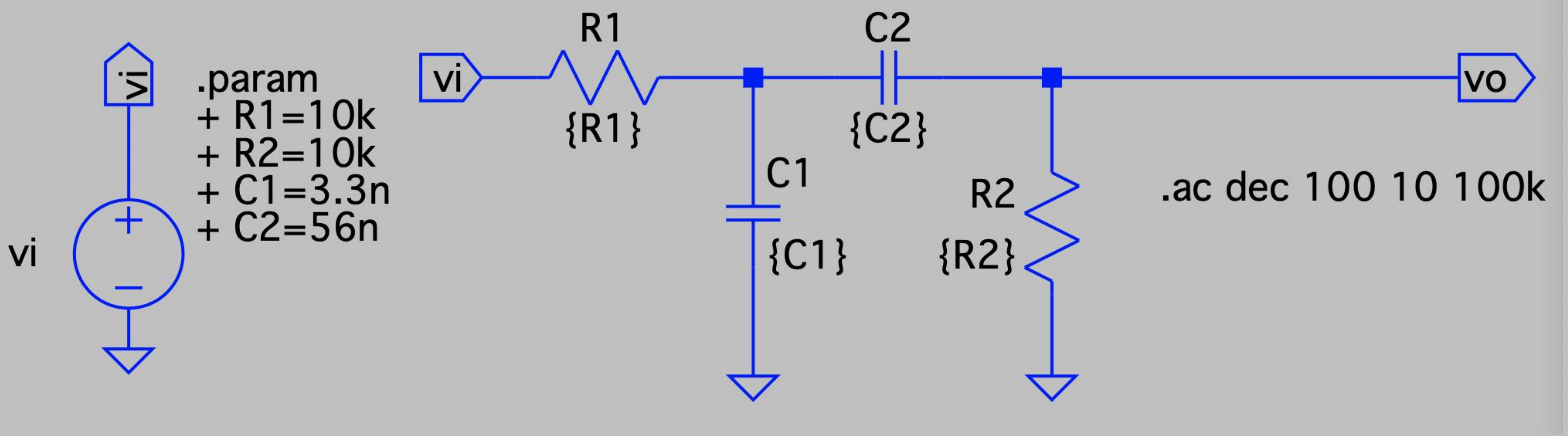
$$A_{v(\text{LF})} = \lim_{s\to 0}\{\tilde{A}_v(s)\} = \frac{0}{G_1 G_2} \quad A_{v(\text{LF})} = 0\ {}^{\text{V}}/_{\text{V}}$$

$$A_{v(HF)} = \lim_{s\to\infty}\{\tilde{A}_v(s)\} = \lim_{s\to\infty}\left\{\frac{G_1}{(C_2)\infty}\right\} = \frac{1}{\infty}\ {}^{\text{V}}/_{\text{V}}$$

$$A_{v(HF)} = 0\ {}^{\text{V}}/_{\text{V}}$$

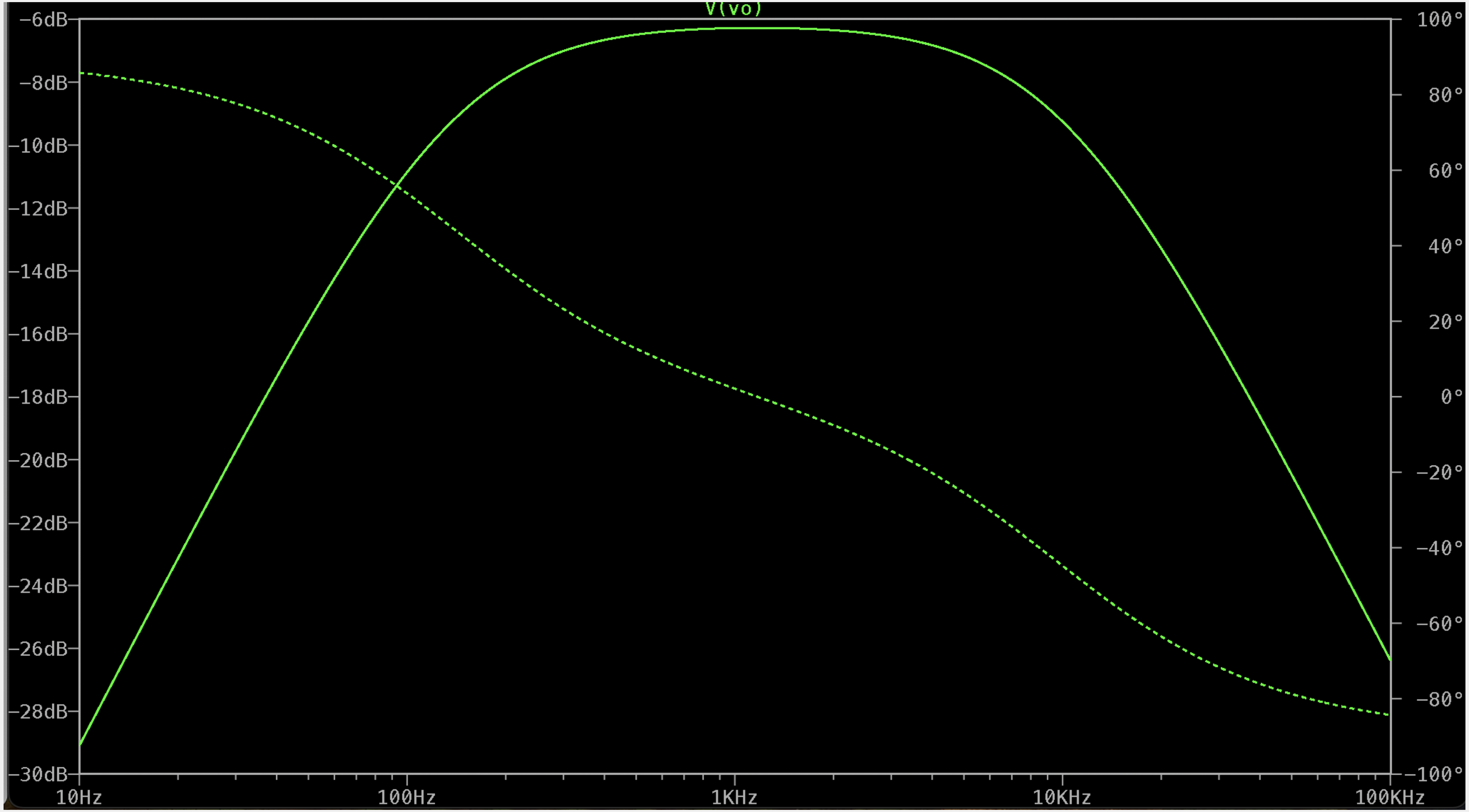
Appendix III — Simulations (LTspice)

Ltspice Description



Appendix III — Simulations (LTspice)

ac Sweep 10 Hz to 100 kHz



Lab 01

Appendix IV — MATLAB

Graph of the Phase of the Gain, $\theta(f)$ (°)

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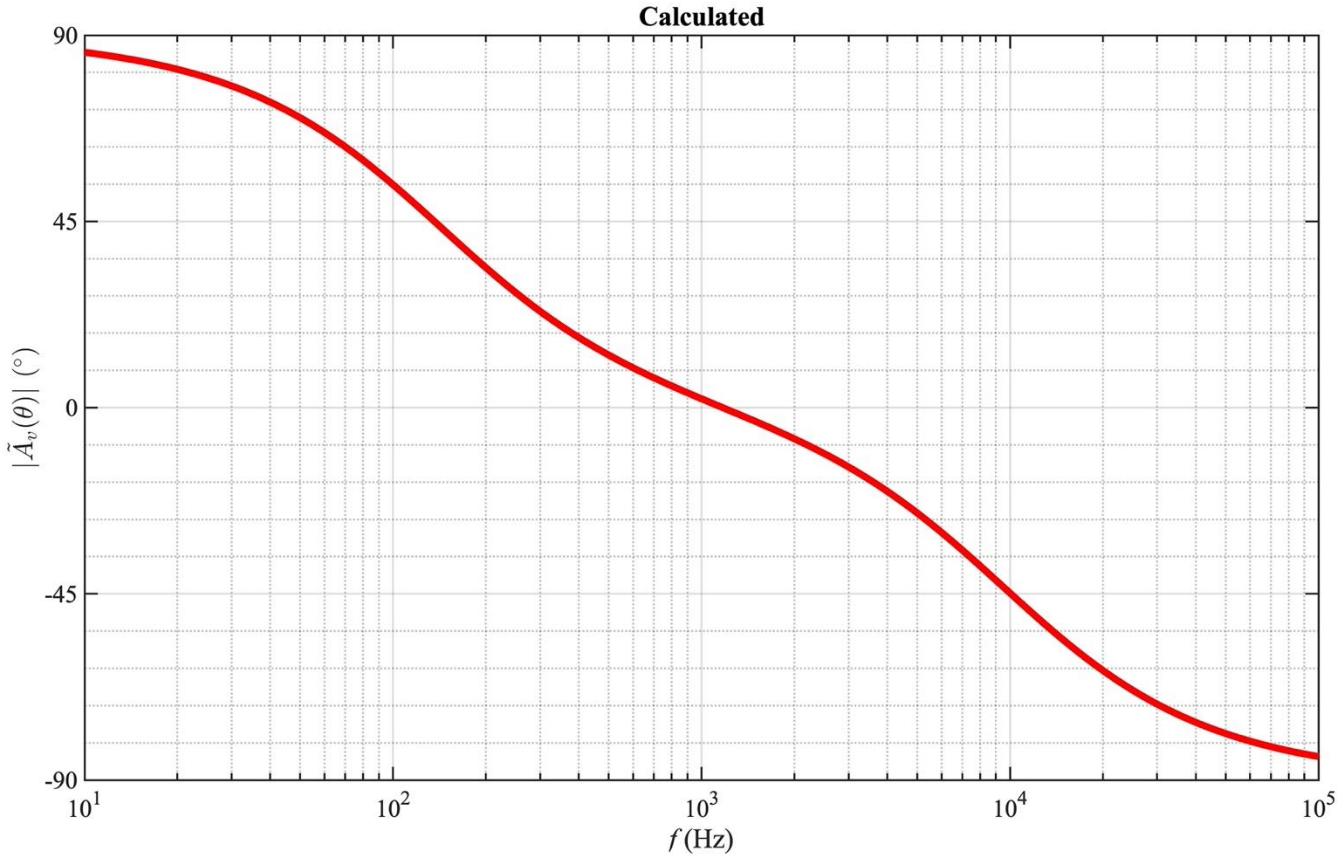

End of Presentation

By Steven Placzek
and Jeremy Burke